EQN

ALGEBRA FACTORING STUDIO & POLYNOMIAL TILES

Kinesthetic Quadratics, Factoring & Polynomial Binomial Arrays

SERIES 1 • CARD 06 / 12 • DOMAIN: POLYNOMIAL GEOMETRY & BINOMIAL FACTORING

CARD 06 / 12 • ALGB-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

GEOMETRIC AREA MODEL. MULTIPLY BY DIMENSION.

Quadratic equations model geometric area. Polynomial tiles model multiplication geometrically:

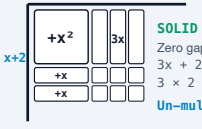
Area = Length × Width = (x + a)(x + b)

Arranging tiles into a rectangle factors polynomials!

CALIBRATED POLYNOMIAL DATUM

- Unit Datum: +1 Constant Tile = 1U = 7.00 mm pitch
- Linear Rod: +x Tile = 5U × 1U = 35.00 × 7.00 mm
- Quadratic: +x² Tile = 5U × 5U = 35.00 × 35.00 mm

AREA MODEL: (x + 3)(x + 2) = x² + 5x + 6



SOLID RECTANGLE

Zero gaps or overlaps

3x + 2x = 5x

3 × 2 = 6 units

Un-multiplying area!

2. KIT PARTS INVENTORY (43 PIECES)

COMPONENT	QTY	SPEC	COLOR
Quadratic (+x²)	2x	38.6×38.6 mm	Matte Blue
Quadratic (−x²)	2x	38.6×38.6 mm	Muted Red
Linear Rod (+x)	6x	38.6×19.0 mm	Matte Blue
Linear Rod (−x)	6x	38.6×19.0 mm	Muted Red
Constant (+1)	12x	19.0×19.0 mm	Matte Blue
Constant (−1)	12x	19.0×19.0 mm	Muted Red
Cartesian Rails	2x	Stage L-Rails	PLA White
3x1 Storage Bin	1x	49.0 mm	PLA White

TOTAL: 40 Polynomial Tiles + 2 L-Rails + 7U Box = 43 Parts

SLANT 3D PRODUCTION PALETTE

- PLA Matte Blue: Positive polynomial values (+x², +x, +1).
- PLA Matte Muted Red: Negative polynomial values (-x², -x, -1).
- PLA White: Storage enclosure body and Cartesian L-rails.
- Zero-Pair Elimination: Blue cancels red on sight.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Lay quadratic tiles flat on the box floor, slot linear rods in side bays, and stack unit cubes in center bays. L-rails snap into lid retention for rattle-free carry.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Every degree-n polynomial has exactly n complex roots counted with multiplicity.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: FACTORING TRINOMIALS & BINOMIAL ARRAYS

Step-by-Step Hands-On Tutorial: Factoring $x^2 + 5x + 6 = (x+2)(x+3)$

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/algebra-boxes/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS HSA.SSE.A.2 (Structure)
- CCSS HSA.APR.A.1 (Arithmetic)
- TEKS 111.39.c.10.E (Factoring Tr...)

Grades 7–11 Pre-Algebra & Algebra II

GUIDING INQUIRY

Can you arrange the linear rods so the rectangle closes completely without any gaps or overlapping edges?

1 STEP 1: SEAT ORIGIN TILE (+x²)

Slide the two L-rails into the lid corner to form the origin.
Place the emerald +x² tile flush into the corner.
Its 35.00 × 35.00 mm area establishes the quadratic core.

2 STEP 2: DISTRIBUTE 5x LINEAR RODS

We need to distribute 5 positive x rods around the +x² tile.
Place 3 rods along the top rail (horizontal dimension).
Place 2 rods along the left rail (vertical dimension).

3–4 STEPS 3 & 4: COMPLETE RECTANGLE & SOLVE

3. COMPLETE THE RECTANGLE: Fill missing corner with 6 constant (+1) tiles!
4. READ BINOMIAL DIMENSIONS: Horizontal width is (x + 3); vertical length is (x + 2).
Total Area: $(x + 2)(x + 3) = x^2 + 5x + 6$!
Calculator Verification: Press Polynomial Solver Icon ([⁻¹][EQN] & [⁻¹][POLY]) and run [⁻¹]
[POLY] 1 [A] 5 [B] 6 [C] 0 [D] [XEQ] [RCL] 1 [RCL] 3 → -2.

5 CORE PRINCIPLE: FACTORING IS UN-MULTIPLYING AREA

Factoring a quadratic is literally finding the length and width of a rectangle! If the tiles cannot form a solid rectangle without gaps, the trinomial is prime. Zero-pairs (+x and -x) let you model differences of squares x^2 - 4 with effortless physical cancellation!

★ EXTENSION INVESTIGATION: FACTOR: x² + 6x + 8

Arrange one +x² tile, six +x rods, and eight +1 units into a solid rectangle. What are its rail dimensions? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.